

Coherent Algebraic Sheaves

Jean-Pierre Serre

Translated by Quentin Asparria and Axel Tamir

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Chapter 1

Sheaves

1.1 Operations on Sheaves

1.1.1 Definition of a sheaf

Let X be a topological space. A *sheaf of abelian groups* on X (or simply a *sheaf*) consists of:

- (a) A function $x \rightarrow \mathcal{F}_x$ sending every $x \in X$ to an abelian group $\mathcal{F}_x$,
- (b) A topology on the set $\mathcal{F}$, the sum of the sets $\mathcal{F}_x$.

If f is an element of $\mathcal{F}_x$, we set $\pi(f) = x$; the map π is called the projection of $\mathcal{F}$ onto X ; the subset of $\mathcal{F} \times \mathcal{F}$ formed by the pairs (f, g) such that $\pi(f) = \pi(g)$ is denoted $\mathcal{F} + \mathcal{F}$.

With these definitions in place, we can state the two axioms to which the data (a) and (b) are subject:

- (I) For all $f \in \mathcal{F}$, there exist open neighborhoods V of f and U of $\pi(f)$ such that the restriction of π at V is a homeomorphism of V onto U . (In other words, if π is a local homeomorphism)
- (II) The mapping $f \rightarrow -f$ is a continuous mapping of $\mathcal{F}$ into $\mathcal{F}$, and the mapping $(f, g) \rightarrow f + g$ is a continuous mapping of $\mathcal{F} + \mathcal{F}$ into $\mathcal{F}$.

It will be observed that, even if X is separated (which we do not assume), $\mathcal{F}$ is not necessarily separated, which is illustrated by the example of germs of functions.

Example of a sheaf. For an abelian group G , let us set $\mathcal{F}_x = G$ for every $x \in X$; the set $\mathcal{F}$ can be identified with the product $X \times G$, and, if we equip it with the product topology of X by the discrete topology of G , we obtain a sheaf, called the *constant sheaf* isomorphic with G , often identified with G .

1.1.2 Sections of a sheaf

Let $\mathcal{F}$ be a sheaf on the space X , and let U be a subset of X . We define a section of $\mathcal{F}$ over U to be a continuous map $s : U \rightarrow \mathcal{F}$ such that $\pi \circ s$ is the identity mapping on U . So we have $s(x) \in \mathcal{F}_x$ for all $x \in U$. The set of sections of $\mathcal{F}$ over U is denoted by $\Gamma(U, \mathcal{F})$; axiom (II) implies that $\Gamma(U, \mathcal{F})$ is an abelian group. If $U \subset V$, and if s is a section over V , the restriction of s at U is a section over U ; hence we have a homomorphism $\rho_U^V : \Gamma(V, \mathcal{F}) \rightarrow \Gamma(U, \mathcal{F})$.

If U is open on X , $s(U)$ is open in $\mathcal{F}$, and, U runs over a base of the topology of X , then $s(U)$ runs over a base of the topology of $\mathcal{F}$; this is only another way of stating axiom (I)...

Note another consequence of axiom (I): For every $f \in \mathcal{F}_x$, there exists a section s over a neighborhood of x such that $s(x) = f$, and two sections enjoying this property coincide on a neighborhood of x . In other words, $\mathcal{F}_x$ is an *inductive limit* of $\Gamma(U, \mathcal{F})$ following the filtering order of the neighborhoods of x .